

Møller Detector Electronics Upgrading

23 August 2006

The main goal of the electronic upgrading for the Møller Polarimeter is to get more higher bandwidth (up to 200MHz) of the detector system. The bottleneck of the current detector electronics is PLU module type of LeCroy-2365, that has bandwidth <75MHz. Another module needed to upgrading is discriminator type of Ortec-TD8000, that has input rate <150MHz (Pic.1).

The proposed electronics will be built using commercially available components which follow NIM, VME or CAMAC standards. To accomplish high single rates in the detector subsystems, only modules capable of handling rates of 150MHz or more are acceptable.

After some searching of the replacement candidates for these units, we founded the next suitable units:

Type	Model	Form Factor	Units	Price*
Programmable Logic Unit 8x8	Caen C542	CAMAC	1	\$4,550
Programmable Logic Unit 8x8	Caen V495	VME-V430	1	\$4,550
General Purpose VME Board (as PLU)	Caen V1495	VME	1	\$3.400
Time to Digital Converter	Caen V1190B	VME	1	\$7.095
Leading Edge Discriminator P/S 708	P/S 708	NIM	2	\$1,500
Quad 300MHz Majority Logic	P/S 754	NIM	2	\$1,500

The short description of the PLU Caen C542 Unit.

Dual Programmable Logic Unit is 1-unit wide CAMAC module housing 2 independent Look-Up Tables with 8 Input and 8 Output channels each. Each Look-Up Table is realized by a 256 locations RAM that can be addressed either via the input signals or via the CAMAC bus; the RAM data bus can be read and written via the CAMAC bus or fed to the output connectors. Each section has three different working modes that can be selected independently:

- Look-Up Table;
- Pattern Generator;
- 8 bit Input/Output Register

The maximal frequency is 200MHz in "Normal" output mode. Front panel connectors for input and output signals are LEMO type. We are most interested in the "Look-Up Table" mode of this unit.

Leading Edge Discriminator 708.

- Guaranteed 300 MHz Operation
- Eight Totally Independent Channels
- Deadtimeless Updating Output Stage

A fifteen turn potentiometer provides continuous output width adjustment from 2nSec to over 50nSec for each channel. The threshold is variable from -10mV to -1Volt with a fifteen turn

potentiometer on each channel. Front panel connectors for input and output signals are LEMO type. For remotely adjustment of the thresholds and output pulse width the discriminator module modification is needed.

Modification of the Leading Edge Discriminator 708.

The one of the possibility for remote adjusting thresholds is using external 8 channels DAC unit with output voltage level from 0 to -10V (Model BiRa 5408 DAC). For connection of the DAC outputs to the discriminator we need to provide input connectors on the module. We can use for this goal connectors on the front panel with inverted outputs (/OUT).

There is detailed steps to modify unit for remotely thresholds adjustment.

1. Cut connection on soldering side of the PCB board between:

- channel 1: R1,R9 and RP1
- channel 2: R2,R10 and RP2
- channel 3: R3,R11 and RP3
- channel 4: R4,R12 and RP4
- channel 5: R5,R13 and RP5
- channel 6: R6,R14 and RP6
- channel 7: R7,R15 and RP7
- channel 8: R8,R16 and RP8

2. Disconnect by unsoldering front panel connector /OUT.

3. Connect connector /OUT throw resistor 100-200 Ohm to:

- channel 1: R1,R9
- channel 2: R2,R10
- channel 3: R3,R11
- channel 4: R4,R12
- channel 5: R5,R13
- channel 6: R6,R14
- channel 7: R7,R15
- channel 8: R8,R16

To remotely adjustment of the output pulse width we suppose to replace installed multi-turn potentiometer with digital one. The most suitable candidate presented on the market today are next:

Type	Manufacturer	Taps	Resistance	Interface
MAX5439	Maxim Integrated Products, Inc.	128	100k	SPI
AD5290	Analog Devices	128	100k	SPI, 3-wire
AD7376	Analog Devices	256	100k	SPI

Digital potentiometers are controlled by using serial 3-wire or SPI interface.

This interface can be implemented with using digital I/O board, for example VME board type of VMICVME2510. There is possible to place digital potentiometers inside of the module.

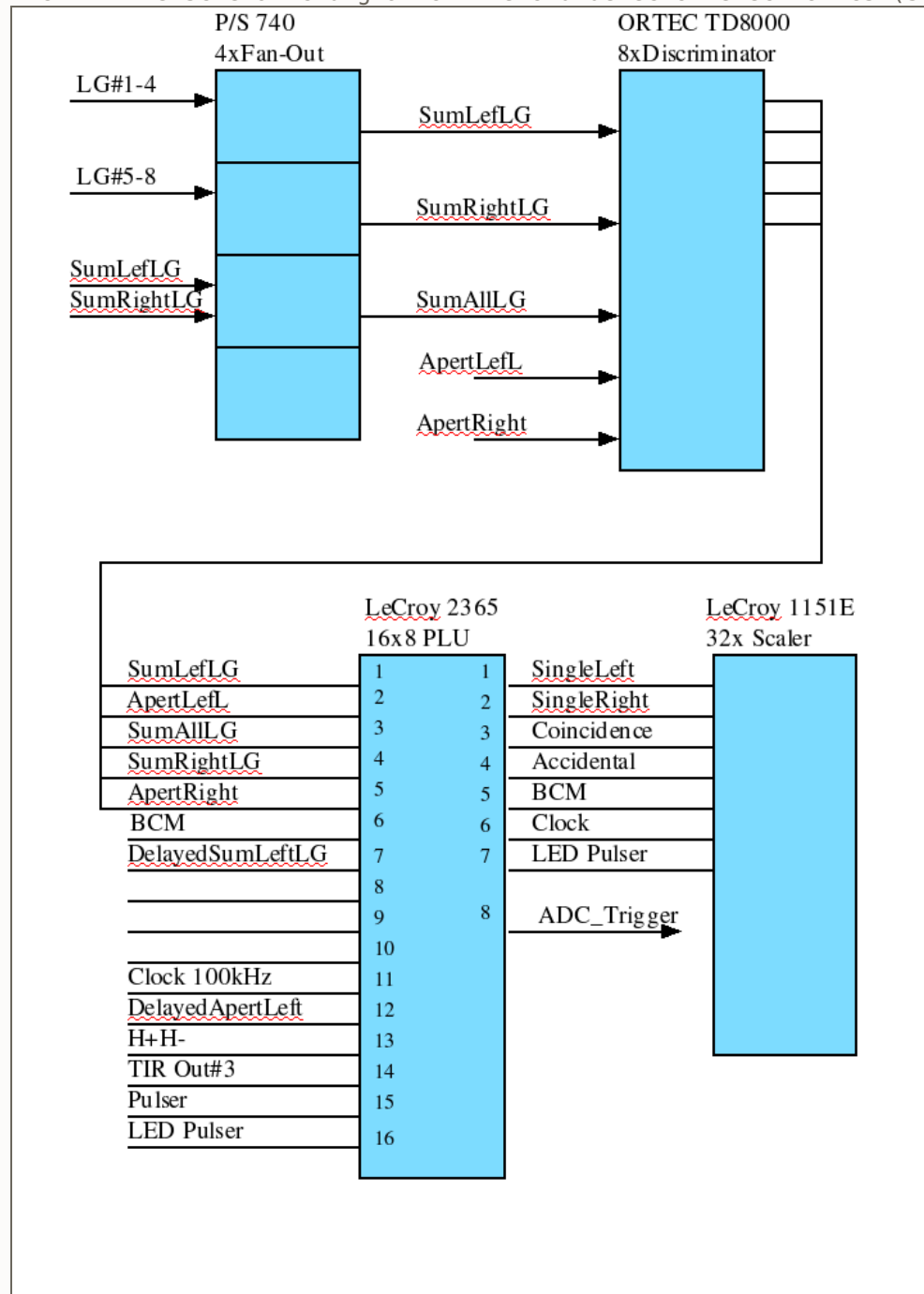
To do this, the connector for SPI interface should be located on the back panel of the unit.

The scheme of the detector electronics after upgrading is shown on Pic.2.

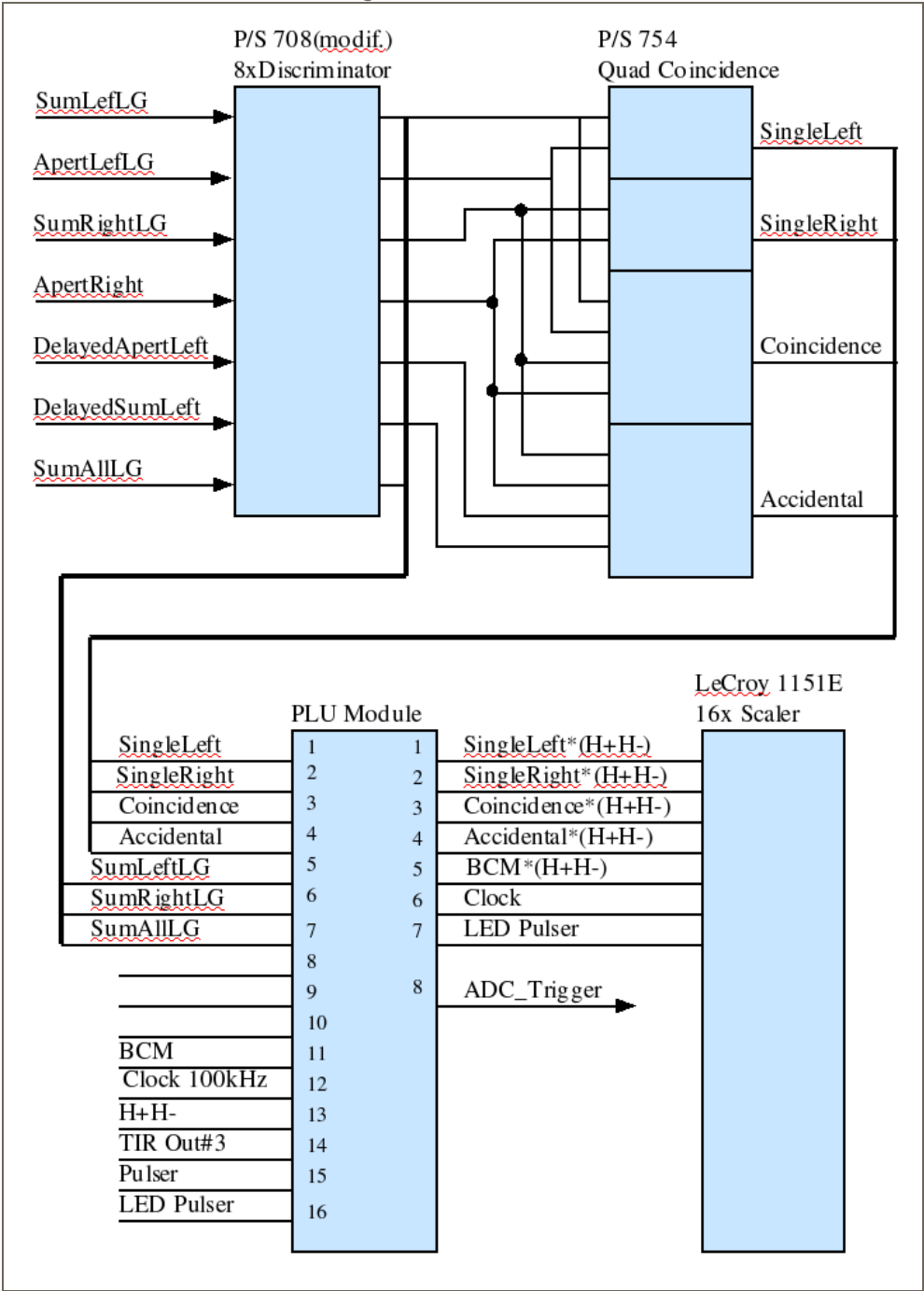
The signals with high rate are splitting from others low rate signals. The coincidence of these high rate signals performed on the Quad Majority Logic Unit type of P/S 754. The output signals from this module going to the PLU inputs.

The example of the signal connections in case of using PLU type of C542(V495) is shown on Pic.3.

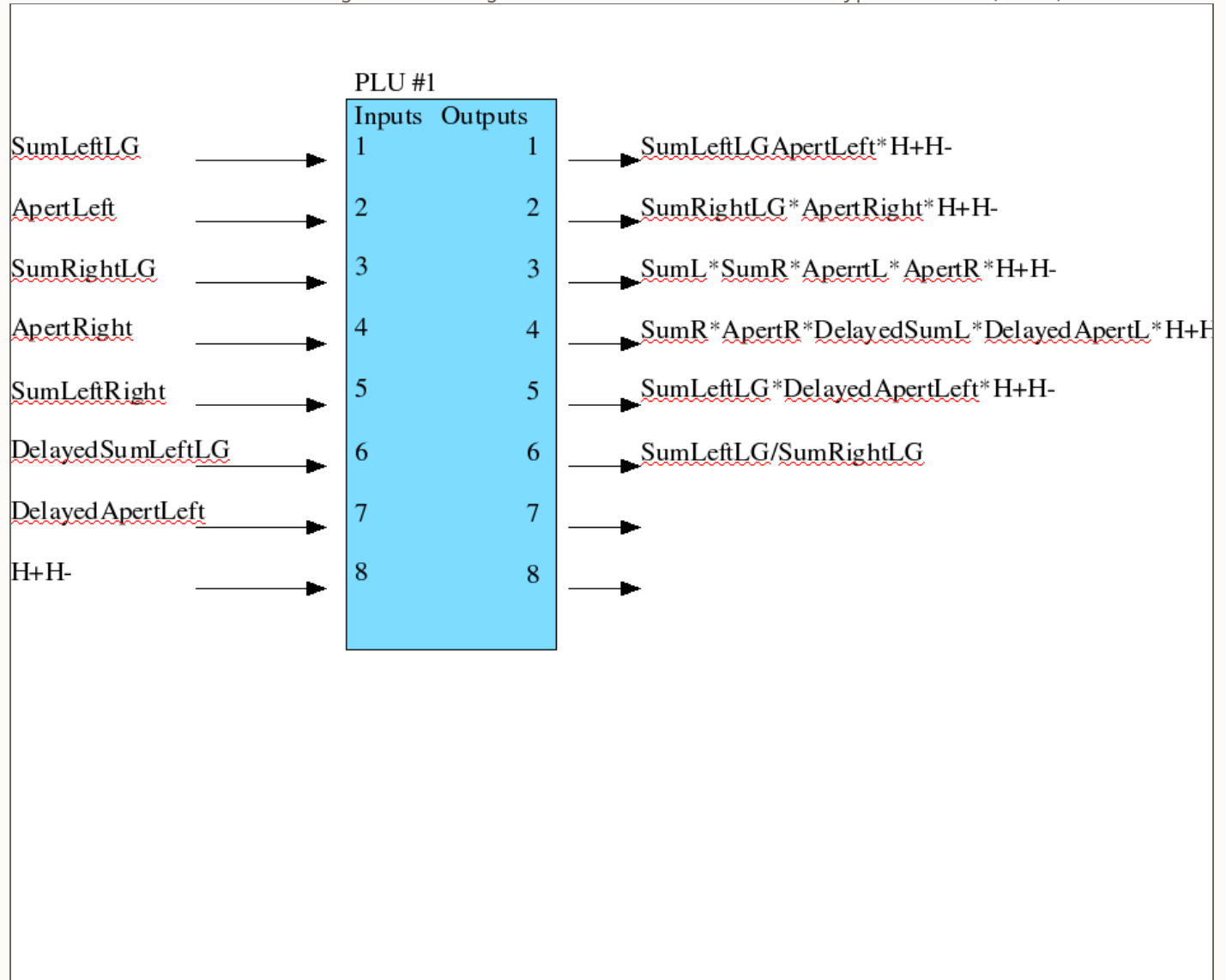
Pic. 1. The schematic digram of the old detector electronics (simplified).



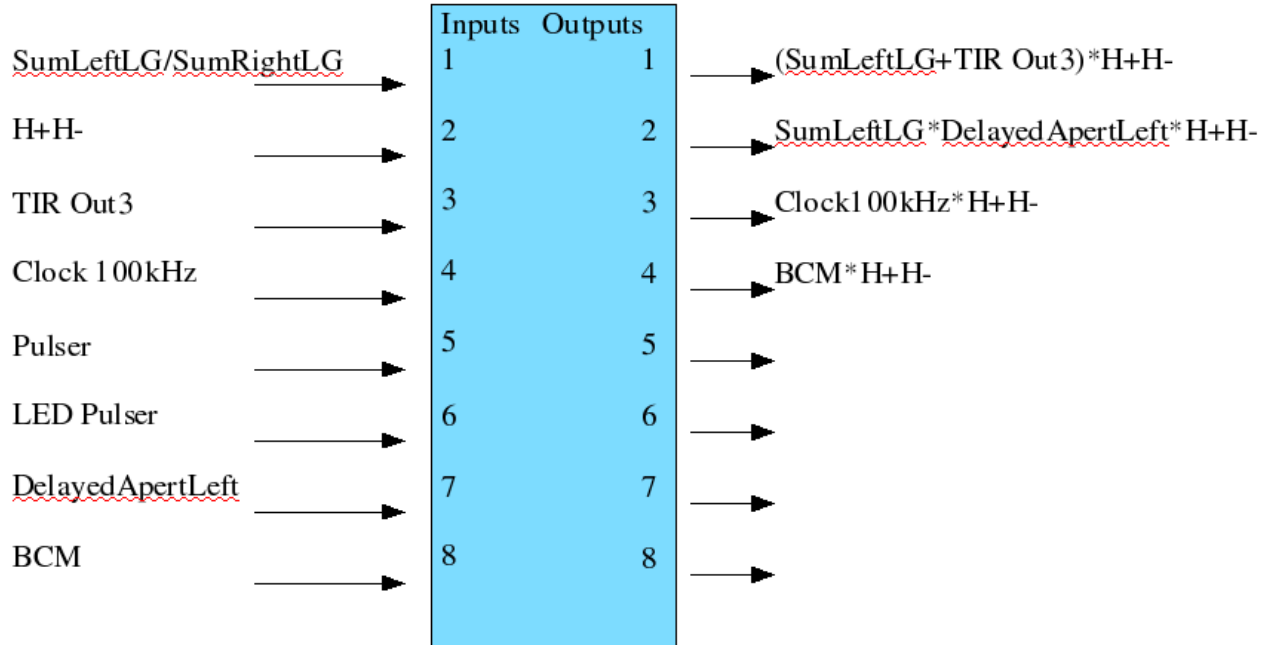
Pic. 2. The schematic diagram of the new detector electronics.



Pic. 3. The schematic diagrams of signals connections for PLU type of C542(V495).



PLU #2



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